# Complete finite orbit certificate for spherical matching extensions

Human Compression Project

This appendix prints the entire required finite program. It constructs the five spherical actions, checks the full Coxeter orders and actual vertex stabilizers, and enumerates every unordered pair. The independent fixed-pair sum is a second count within the same exact GAP execution. The classical finite Coxeter orders and GAP's group algorithms are explicit trusted inputs; a small unexplained permutation quotient would not prove the source identification. The output supplies a finite premise of the uniform written proof, not evidence obtained by sampling the parameter $s$.

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# Reconstruct every exceptional spherical action from its literal Coxeter
# presentation, then enumerate ALL unordered pairs. No source orbit table is read.
SizeScreen([100000,24]);;
Check278:=function(t,m) if not t then Error("PHASE278_REJECT ",m);fi;end;;
rows278:=[
 [[3,5],120,12,5,[1,3,5],[30,30,6]],
 [[5,3],120,20,5,[1,2,3,4,5],[30,60,60,30,10]],
 [[3,4,3],1152,24,6,[1,3,4,6],[96,72,96,12]],
 [[3,3,5],14400,120,15,[1,4,6,7,9,10,12,15],[720,1200,720,1800,720,1200,720,60]],
 [[5,3,3],14400,600,15,[1..15],[]]
];;
# Short words beta^a t_j beta^b with ZERO-based source j, and unordered orbit size.
missing278:=[[-3,3,-1,3600],[-4,0,-2,1200],[-4,3,-1,7200],
 [-5,0,-2,2400],[-5,3,-1,7200],[-4,3,-2,3600],[-6,3,-1,7200],
 [-9,1,4,1200],[-5,3,-2,7200],[-7,3,-1,14400],[-5,3,-3,1800],
 [-8,3,-1,7200],[-7,3,-2,7200],[-6,3,-4,1200],[-9,3,-1,3600],
 [-8,3,-2,7200],[-7,3,-4,2400],[-10,3,-1,7200],[-7,3,-5,1200],[-10,3,-2,3600]];;
Census278:=function(row)
 local labels,n,F,r,rels,i,j,Q,q,ct,t,G,H,beta,alpha,h,vcount,pointorbs,
  pairs,orbs,orbitof,ix,z,oldids,missids,m,w,bad,allids,sz,pair,vertices,burnside,g,fix;
 labels:=row[1];n:=Length(labels)+1; F:=FreeGroup(n);r:=GeneratorsOfGroup(F);
 rels:=List(r,x->x^2);
 for i in [1..n] do for j in [i+1..n] do
  if j=i+1 then Add(rels,(r[i]*r[j])^labels[i]);else Add(rels,(r[i]*r[j])^2);fi;
 od;od;
 Q:=F/rels;q:=GeneratorsOfGroup(Q);ct:=CosetTable(Q,Subgroup(Q,q{[2..n]}));
 t:=List([1..n],i->PermList(ct[2*i-1]));G:=Group(t);H:=Group(t{[2..n]});
 vcount:=Length(ct[1]);vertices:=[1..vcount];
 Check278(Size(G)=row[2] and vcount=row[3],"spherical action order and degree");
 Check278(Set(Orbit(G,1))=vertices,"transitive action");
 Check278(H=Stabilizer(G,1),"actual vertex stabilizer");
 Check278(Size(H)*vcount=Size(G),"vertex action index");
 beta:=Product(t);h:=row[4];alpha:=beta^h;
 Check278(Order(beta)=2*h and Order(alpha)=2,"Coxeter element and antipode orders");
 Check278(ForAll(t,x->Comm(x,alpha)=()) and ForAll(vertices,x->x^alpha<>x),"central fixed-point-free antipode");
 pointorbs:=OrbitsDomain(H,vertices);
 pairs:=Combinations(vertices,2);orbs:=OrbitsDomain(G,pairs,OnSets);
 Check278(Union(orbs)=pairs and Sum(List(orbs,Length))=Binomial(vcount,2),"complete disjoint unordered pair
  ↪  census");
 # A second count uses the action on all elements and the fixed-pair formula.
 burnside:=0;
 for g in Elements(G) do
  fix:=vcount-NrMovedPoints(g);
  burnside:=burnside+Binomial(fix,2)+Number(CycleLengths(g,vertices),x->x=2);
 od;
 Check278(burnside=Length(orbs)*Size(G),"independent fixed-pair Burnside count");
 # For every orbit record all its representatives whose first endpoint is1.
 orbitof:=[];
 for ix in [1..Length(orbs)] do for pair in orbs[ix] do
  if 1 in pair then z:=First(pair,x->x<>1);Check278(not IsBound(orbitof[z]),"disjoint orbit
   ↪  assignment");orbitof[z]:=ix;fi;
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 od;od;
 Check278(ForAll([2..vcount],i->IsBound(orbitof[i])),"all basepoint pairs assigned");
 oldids:=List(row[5],i->orbitof[1^(beta^i)]);
 Check278(Length(Set(oldids))=Length(oldids),"distinct old representatives");
 Check278(Length(orbs[Last(oldids)])=vcount/2 and ForAll(orbs[Last(oldids)],p->p[1]^alpha=p[2]),"antipodal
  ↪  matching orbit");
 if labels<>[5,3,3] then
  Check278(Set(oldids)=[1..Length(orbs)],"source representative list complete");
  Check278(List(oldids,i->Length(orbs[i]))=row[6],"exact source-ordered orbit sizes");
 else
  Check278(Length(pointorbs)=45 and Length(orbs)=35,"120-cell full census is35");
  Check278(Length(oldids)=15,"literal source list has15");
  missids:=[];
  for m in missing278 do
   w:=beta^m[1]*t[m[2]+1]*beta^m[3];z:=1^w;
   Check278(z<>1,"missing word moves basepoint");ix:=orbitof[z];
   Check278(not ix in oldids and not ix in missids,"missing orbit new and distinct");
   Check278(Length(orbs[ix])=m[4],"missing representative orbit size");Add(missids,ix);
  od;
  Check278(Length(missids)=20 and Union(oldids,missids)=[1..Length(orbs)],"corrected35 representatives
   ↪  exhaustive");
  Check278(Collected(List(orbs,Length))=[[300,1],[1200,6],[1800,1],[2400,2],[3600,6],[7200,18],[14400,1]],"ful
   ↪  l120-cell size multiplicities");
  Print("PHASE278_MISSING_WORDS ",missing278,"\n");
 fi;
 Print("PHASE278_ROW
  ↪  {\"labels\":",labels,",\"order\":",Size(G),",\"vertices\":",vcount,",\"stabilizer_order\":",Size(H),",\"c
  ↪  oxeter_half_order\":",h,",\"point_orbit_sizes\":",List(pointorbs,Length),",\"unordered_orbit_sizes\":",Li
  ↪  st(orbs,Length),",\"burnside_numerator\":",burnside,",\"old_exponents\":",List(row[5],i->i),",\"old_orbit
  ↪  _sizes\":",List(oldids,i->Length(orbs[i])),",\"missing_orbits\":",Length(orbs)-Length(oldids),"}\n");
end;;
for row278 in rows278 do Census278(row278);od;
Print("PHASE278_EXCEPTIONAL_CENSUS_PASS five_actions all_pairs exact35_with20_missing\n");
QUIT;
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